

# Aerial-Ground Person Re-Identification

## Leaderboards

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**Abstract.** This document presents leaderboard tables summarizing reported experimental results on five Aerial-Ground Re-Identification (AG-ReID) benchmarks. All results are collected from **formally published papers**. Therefore, Results reported only in arXiv preprints are therefore not included.

**Keywords:** Aerial-ground person re-identification · Leaderboards

### Overview

The leaderboard tables are designed to provide a **concise, visually clear and publication-ready** comparison format. **Each table reserves the final row for users to insert their own experimental results**, and the best result under each evaluation metric is highlighted in **bold**. For readability, the reported methods in each table are divided into three groups: representative CNN-based methods, representative ViT-based methods, and methods specifically designed for AG-ReID.

### Usage

The file `ag_reid_leaderboards.tex` in the current directory is the L<sup>A</sup>T<sub>E</sub>X source of `ag_reid_leaderboards.pdf`. Users may directly copy the corresponding tables from this source file for use in their own manuscripts.

**Table 1: Performance comparison on AG-ReID.** Best results are in **bold**.

| Method            | A→G          |              | G→A          |              |
|-------------------|--------------|--------------|--------------|--------------|
|                   | mAP          | R1           | mAP          | R1           |
| BoT [12]          | 55.47        | 70.01        | 58.83        | 71.20        |
| OsNet [29]        | 58.32        | 72.59        | 60.99        | 74.22        |
| SBS [3]           | 59.77        | 73.54        | 62.27        | 73.70        |
| VV [6]            | 67.23        | 77.22        | 69.83        | 79.73        |
| ViT [2]           | 72.61        | 81.47        | 73.39        | 82.85        |
| TransReID [4]     | 73.23        | 81.33        | 75.71        | 84.10        |
| CLIP-ReID [9]     | 75.68        | 82.59        | 78.51        | 85.68        |
| Explain [14]      | 72.38        | 81.28        | 73.35        | 82.64        |
| VDT [28]          | 74.44        | 82.91        | 78.57        | 86.59        |
| DTST [20]         | 74.51        | 83.48        | 76.05        | 84.10        |
| GSAIign [8]       | 75.01        | 83.75        | 77.73        | 84.10        |
| VIF [5]           | 75.22        | 83.75        | 79.19        | 87.32        |
| SD-ReID [21]      | 75.40        | 85.16        | 77.02        | 85.57        |
| SAS-VPReID [24]   | 75.98        | 84.24        | 77.28        | 86.17        |
| SeCap [19]        | 76.16        | 84.03        | 78.34        | 87.01        |
| LATex [26]        | 77.67        | 85.26        | 81.15        | 89.40        |
| H <sup>2</sup> RL | 79.21        | 86.06        | 81.28        | 87.94        |
| CVAf [13]         | 79.45        | 87.23        | 82.59        | 90.12        |
| GLPSG [30]        | <b>81.50</b> | <b>87.70</b> | <b>84.30</b> | <b>90.60</b> |
| <b>Yours</b>      |              |              |              |              |

**Table 2: Performance comparison on AG-ReID v2.** Best results are in **bold**.

| Method            | A→C          |              | C→A          |              | A→W          |              | W→A          |              |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   | mAP          | R1           | mAP          | R1           | mAP          | R1           | mAP          | R1           |
| Swin [11]         | 57.66        | 68.76        | 57.70        | 68.80        | 56.15        | 68.49        | 53.90        | 64.40        |
| HRNet-18 [18]     | 65.07        | 75.21        | 66.16        | 76.25        | 66.17        | 76.26        | 66.17        | 76.25        |
| SwinV2 [10]       | 66.09        | 76.44        | 62.14        | 77.11        | 69.09        | 80.08        | 65.61        | 74.53        |
| MGN(R50) [17]     | 70.17        | 82.09        | 72.41        | 84.21        | 78.66        | 88.14        | 73.73        | 84.06        |
| BoT(R50) [12]     | 71.49        | 80.73        | 69.67        | 79.46        | 75.98        | 86.06        | 72.41        | 82.69        |
| SBS(R50) [3]      | 72.04        | 81.96        | 73.89        | 84.10        | 78.94        | 88.14        | 75.01        | 84.66        |
| PCL-CLIP [7]      | 72.20        | 79.80        | 72.40        | 81.12        | 77.70        | 87.14        | 73.89        | 84.19        |
| BoT(ViT) [12]     | 77.03        | 85.40        | 75.90        | 84.65        | 80.48        | 89.77        | 75.90        | 84.65        |
| ViT [2]           | 77.03        | 85.40        | 79.30        | <b>89.77</b> | 80.48        | 89.77        | 77.42        | 88.24        |
| CLIP-ReID [9]     | 79.79        | 85.36        | 79.08        | 85.64        | 84.23        | 89.14        | 79.55        | 86.50        |
| FusionReID [22]   | 80.70        | 86.70        | 80.00        | 87.90        | 84.20        | 89.70        | 80.90        | 86.50        |
| TransReID [4]     | 81.40        | 88.00        | 80.10        | 87.60        | 84.50        | 90.40        | 81.10        | 87.70        |
| SAS-VPReID [24]   | 76.39        | 83.40        | 78.47        | 85.26        | 82.10        | 89.45        | 78.47        | 85.73        |
| Explain [14]      | 79.00        | 87.70        | 78.24        | 87.35        | 83.14        | <b>93.67</b> | 79.08        | 87.73        |
| VDT [28]          | 79.13        | 86.46        | 78.12        | 86.14        | 82.21        | 90.00        | 78.52        | 85.26        |
| SD-ReID [21]      | 80.61        | 87.04        | 79.24        | 86.74        | 84.06        | 90.86        | 80.12        | 86.79        |
| V2E [15]          | 80.72        | 88.77        | 78.51        | 87.86        | 84.85        | 93.62        | 80.11        | 88.61        |
| SeCap [19]        | 80.84        | 88.12        | 79.99        | 88.24        | 84.01        | 91.44        | 80.15        | 87.56        |
| GSAIign [8]       | 81.38        | 87.86        | 81.05        | 88.02        | 83.98        | 90.63        | 80.90        | 87.31        |
| CVAf [13]         | 83.35        | 88.77        | 83.17        | 88.40        | 87.81        | 92.49        | <b>85.08</b> | <b>90.85</b> |
| LATex [26]        | 83.50        | 89.13        | 82.85        | 89.01        | 86.35        | 91.35        | 83.30        | 89.32        |
| ViSA [27]         | 83.61        | 89.43        | 82.32        | 88.57        | 85.99        | 91.63        | 83.10        | 89.23        |
| H <sup>2</sup> RL | <b>84.31</b> | <b>89.90</b> | <b>83.55</b> | 88.74        | <b>88.03</b> | 92.76        | 84.42        | 89.83        |
| <b>Yours</b>      |              |              |              |              |              |              |              |              |

**Table 3: Performance comparison on CARGO.** Best results are in **bold**.

| Method            | ALL          |              | A→G          |              | A→A          |              | G→G          |              |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   | mAP          | R1           | mAP          | R1           | mAP          | R1           | mAP          | R1           |
| BAU [1]           | 38.40        | 45.20        | 36.70        | 40.40        | 42.60        | 50.00        | 51.20        | 61.60        |
| VV [6]            | 38.44        | 45.83        | 29.00        | 31.25        | 49.73        | 67.50        | 62.99        | 72.31        |
| SBS [3]           | 43.09        | 50.32        | 29.00        | 31.25        | 49.73        | 67.50        | 62.99        | 72.31        |
| PCB [16]          | 44.50        | 51.00        | 30.40        | 34.40        | 44.60        | 55.00        | 67.60        | 74.10        |
| BoT [12]          | 46.49        | 54.81        | 32.56        | 36.25        | 49.79        | 65.00        | 66.47        | 77.68        |
| MGN [17]          | 49.08        | 54.81        | 33.47        | 31.87        | 52.96        | 65.00        | 71.05        | 83.93        |
| AGW [25]          | 53.44        | 60.26        | 40.90        | 43.57        | 56.48        | 67.50        | 71.66        | 81.25        |
| ViT [2]           | 53.54        | 61.54        | 40.11        | 43.13        | 64.47        | 80.00        | 71.34        | 82.14        |
| PCL-CLIP [7]      | 60.93        | 67.31        | 51.43        | 54.37        | 60.75        | 70.00        | 76.00        | 84.82        |
| CLIP-ReID [9]     | 64.25        | 68.27        | 53.83        | 55.62        | 65.42        | 75.00        | 80.80        | 84.82        |
| VDT [28]          | 55.20        | 64.10        | 42.76        | 48.12        | 66.83        | <b>82.50</b> | 71.59        | 82.14        |
| DTST [20]         | 55.73        | 64.42        | 43.49        | 50.53        | 63.31        | 80.00        | 72.40        | 78.57        |
| SeCap [19]        | 56.89        | 64.72        | 46.37        | 48.75        | 66.90        | <b>82.50</b> | 75.24        | 82.54        |
| VIF [5]           | 57.46        | 72.76        | 44.55        | 51.25        | 66.98        | <b>82.50</b> | 74.19        | 83.93        |
| SD-ReID [21]      | 57.47        | 65.06        | 46.44        | 53.12        | 67.70        | <b>82.50</b> | 74.08        | 81.25        |
| GSAIign [8]       | 57.95        | 65.06        | 61.55        | 64.89        | 65.55        | 80.00        | 73.86        | 83.04        |
| SAS-VPReID [24]   | 58.48        | 64.42        | 57.36        | 58.51        | 63.41        | 70.00        | 76.90        | 83.93        |
| ViSA [27]         | 65.46        | 70.51        | 69.00        | 71.28        | 67.78        | <b>82.50</b> | <b>83.90</b> | 88.39        |
| LATex [26]        | 67.09        | <b>76.96</b> | 58.88        | 66.87        | 69.06        | 80.00        | 79.90        | <b>90.18</b> |
| H <sup>2</sup> RL | <b>67.65</b> | 74.68        | <b>69.77</b> | <b>75.50</b> | <b>69.60</b> | 75.00        | 82.07        | 88.39        |
| <b>Yours</b>      |              |              |              |              |              |              |              |              |

**Table 4: Performance comparison on LAGPeR.** Best results are in **bold**.

| Method            | A→G          |              | G→A          |              | G→AG         |              |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   | mAP          | R1           | mAP          | R1           | mAP          | R1           |
| CLIP-ReID [9]     | 17.60        | 24.40        | 20.10        | 21.30        | 10.20        | 12.30        |
| ViT [2]           | 27.25        | 38.67        | 30.69        | 32.04        | 15.31        | 18.88        |
| TransReID [4]     | 28.80        | 38.80        | 32.10        | 33.00        | 18.80        | 22.90        |
| MIP [23]          | 29.30        | 39.30        | 32.60        | 33.90        | 17.30        | 21.00        |
| Explain [14]      | 28.89        | 40.48        | 31.91        | 32.96        | 18.91        | 22.33        |
| VDT [28]          | 28.97        | 40.15        | 31.91        | 33.55        | 16.45        | 19.50        |
| SD-ReID [21]      | 29.72        | 40.15        | 32.86        | 34.47        | 18.88        | 22.85        |
| SeCap [19]        | 30.37        | 41.79        | 33.42        | 35.26        | 19.24        | 24.39        |
| ViSA [27]         | 30.42        | 41.37        | 33.59        | 35.06        | 20.31        | 25.61        |
| SAS-VPReID [24]   | 30.76        | 43.83        | 33.40        | 35.42        | 23.02        | 31.12        |
| H <sup>2</sup> RL | <b>33.36</b> | <b>44.68</b> | <b>37.10</b> | <b>39.86</b> | <b>26.72</b> | <b>34.57</b> |
| <b>Yours</b>      |              |              |              |              |              |              |

**Table 5: Performance comparison on CP2108.** Best results are in **bold**.

| Method          | ALL          |              | A→G          |              | G→A          |              |
|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                 | mAP          | R1           | mAP          | R1           | mAP          | R1           |
| SBS [3]         | 17.29        | 27.96        | 23.05        | 36.06        | 23.83        | 39.14        |
| MGN [17]        | 23.93        | 32.45        | 34.02        | 52.27        | 34.26        | 52.58        |
| BoT [12]        | 28.35        | 36.27        | 38.60        | 57.76        | 37.72        | 55.64        |
| ViT [2]         | 37.92        | 49.36        | 42.03        | 61.14        | 44.58        | 61.97        |
| TransReID [4]   | 38.18        | 50.32        | 45.62        | 62.03        | 45.00        | 62.28        |
| CLIP-ReID [9]   | 41.12        | 57.90        | 45.35        | 62.16        | 45.32        | 62.97        |
| VDT [28]        | 33.10        | 46.88        | 33.64        | 42.76        | 31.74        | 40.59        |
| SeCap [19]      | 38.74        | 53.90        | 46.95        | 63.55        | 46.19        | 63.39        |
| SAS-VPReID [24] | <b>42.40</b> | <b>59.23</b> | <b>47.72</b> | <b>64.25</b> | <b>47.04</b> | <b>63.76</b> |
| <b>Yours</b>    |              |              |              |              |              |              |

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